Multinomial & Ordinal Regression

Standard Level: Statistical Methods

1. Method Description

Extends binary logistic regression to 3+ outcome categories.

Multinomial (nominal DV): k-1 logit models vs reference. RRR = exp(beta).

Ordinal (ordered DV): Proportional Odds Model (POM). Assumes same beta across thresholds. Brant test.

2. Examples

Economics: Transport mode choice (car/train/bus/bike). RRR for train vs car = 0.65 per 10 min travel time.

Medicine: Pain intensity (none/mild/moderate/severe). POM, OR = 1.8 for treatment. Brant p=0.23 -> OK.

Social Sciences: Vote intention (party A/B/C/D). Multinomial logit with ref A.

3. Assumptions

Multinomial: IIA (Hausman-McFadden test)
Ordinal: Proportional odds (Brant test)

4. Code Examples

R:

```
multinom(y ~ x1+x2, data=df) # nnet
polr(y ~ x1+x2, data=df, Hess=TRUE) # MASS
```

Python:

```
LogisticRegression(multi_class='multinomial')
OrderedModel(y, X) # statsmodels
```

5. Exercises

1. Party preference A/B/C with age and education. RRR for party B = 0.92 per year age. Interpret. IIA test?

2. School grades (1-6), study time and gender. POM. Brant p=0.04. Meaning? Alternative?